

of time). While Google claimed that its 53-qubit quantum processor reached 'quantum supremacy', and it would take 10,000 years for a state-of-art supercomputer to perform an equivalent task, many have disputed the claim. Still, it was recorded as a significant feat in quantum computing history. Apart from this, there has been substantial development in quantum information processing using quantum machine learning algorithms. Many programming libraries with popular programming languages have emerged, like the Qiskit library for the Python platform and the Q# programming language.

▪ What can be done?

Research in this area is at the very initial stages and hence there's a lot that needs to be discovered and proved with actual results. Undoubtedly, quantum computing is the future of computing. With all the tech giants,

including IBM, Google and Microsoft, investing in R&D in this field, research and job opportunities are sure to flourish in the future.

Quantum computing is catching the attention of young researchers from across the world. Principle-wise, any computational problem that is classically solvable is also quantum solvable. Alternatively, quantum computing respects the Church-Turing Thesis, according to which any computational problem that can be solved by a classical computer is also

solvable by a quantum computer. This means quantum computing offers no additional advantage over traditional computing when it comes to computability, but it does empower us in terms of time complexity for specific problems. Tapping into those avenues where we can use this power for humanity's benefit is the real challenge. Many fascinating concepts and technical aspects still need to be pondered over and significant research needs to be performed in this field. **END** 🐧

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Figure 2: Global edge computing market 2019-2023 (Credit: www.technavio.com)

Cloud computing enhances accessibility, provides virtual storage space, solves backup problems, offers security against unauthorised access and data loss. Cloud computing is not only helping the businesses to adopt trending technologies like AI, ML, advanced analytics and immersive media, but also complementing the seamless adoption of SaaS, IaaS and PaaS technologies.

According to Forrester, the overall edge computing market size is projected to increase from 2.8 billion dollars in 2019 to 9.0 billion dollars by 2024, with a CAGR of 26.5 per cent during the forecast period. In this context, the Meticulous Research report predicts that the overall market of edge computing would augment at a CAGR of 34.9 per cent from 2019 to reach 28.07 billion dollars

by 2027, while cloud computing is going to be more important among the enterprise world.

By moving analytics to the edge, organisations will get more immediate insights and results from IoT devices and sensors. The concepts of edge computing models and business digitisation are going to become a key component for many IoT initiatives. Both the technologies can be leveraged together symbiotically to create a more balanced and efficient technology business model. **END** 🐧

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